ECON 609 - Problem Set 4
Fall 2022

Be sure to include all the material that shows your work and to present your results in a way that is easy to understand.

Consider an individual with preference

$$u(c) = \mathbb{E}_0 \sum_{t=0}^{T} \beta^t U(c_t)$$

with $\beta = \frac{1}{1+\rho} \in (0,1)$. The individual has stochastic income process $\{y_t\}_{t=0}^T$ where $y_t \in Y = \{y_1, \ldots, y_N\}$. The income process is Markov. Let $\pi(y'|y)$ denote the probability that tomorrow's endowment takes the value y' if today's endowment takes the value y. The agents' budget constraint at period t reads as

$$c_t + a_{t+1} = y_t + (1+r)a_t$$

Suppose that the interest rate r is exogenously given, so that r is a parameter. In order to make the problem computable, we first have to specify the parameters of the model.

We will assume that preference can be represented by a period utility function that is of CRRA form

$$U(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}$$

with $\sigma > 0$.

With respect to the time discount rate, let us assume that agents, in the presence of certainty equivalence, like declining consumption profiles, i.e., assume that $\rho > r$. Structure your code in such a way that adjusting (ρ, r) is easy.

With respect to the income process, we consider a discretized version of a simple AR(1) process. In particular, suppose log-income follows a process

$$\log(y_{t+1}) = \delta \log(y_t) + (1-\delta^2)^{\frac{1}{2}} \epsilon_t$$

where the persistence parameter $\delta \in [0,1)$ and ϵ_t is normally distributed with zero mean and variance σ_y^2. Note that for this process

$$Var(\log(y_t)) = \sigma_y^2$$
$$Corr(\log(y_{t+1}), \log(y_t)) = \frac{Cov(\log(y_{t+1}), \log(y_t))}{\sqrt{Var(\log(y_{t+1}) * Var(\log(y_t))}} = \delta.$$

Discretize this process into an N-state Markov chain using Tauchen's procedure. Ideally choose $N \geq 5$. Note that once you have a Markov chain for $\log(y)$, it is straightforward to exponentiate the states to obtain a Markov chain for income levels y.

If you can't manage this, choose $N = 2$ and let income levels and the transition matrix be given by $Y = \{1 - \sigma_y, 1 + \sigma_y\}$ and

$$\pi = \begin{bmatrix} \frac{1+\delta}{2} & \frac{1-\delta}{2} \\ \frac{1-\delta}{2} & \frac{1+\delta}{2} \end{bmatrix}$$

Parameter values for δ, σ_y will be given below.

And we consider a borrowing constraint $a_{t+1} \geq -\bar{A}$, where $\bar{A}$ is a parameter. The initial asset position of the household is $a_0 = 0$ and the initial income realization is $y_0 = y_l$, where y_l is the lowest realization the Markov chain can take.

(1) Recursive Formulation

Formulate the problem of the agent recursively, i.e., write down Bellman equation and derive the stochastic Euler equation.

(2) Computation with Infinite Horizon

For $T = \infty$, write down a computer program that computes the value function $v(a, y)$ and policy functions $a'(a, y)$ and $c(a, y)$ for given choice of the utility functions as well as given parameterization of the income process. Let $\sigma = 2, \delta = 0.8, \sigma_y = 0.2, \bar{A} = 0, \rho = 0.04, r = 0.02$. Plot the consumption function. Simulate one consumption path for 60 periods.

(3) Change of Parameters

How do the policy functions and simulations change with the size σ_y of income risk, its persistence δ, and risk aversion σ? Interpret your result.

(4) General Equilibrium

Incorporate your code from before for $T = \infty$ into general equilibrium to compute stationary equilibria. The algorithm goes as follows:

(a) Guess an interest rate $r \in (0, \rho)$

(b) Use the first order conditions for the firm to determine $K(r)$.

(c) Solve the household problem for given r.

(d) Use the optimal decision rule $a'(a, y)$ together with the exogenous Markov chain π to find an invariant distribution Φ associated with $a'(a, y)$ and π

(e) Compute $Ea(r) = \int a'(a, y) d\Phi_r$

(f) Compute $d(r) = K(r) - Ea(r)$. If $d(r)$ is close to 0, you have found a stationary recursive equilibrium, otherwise, update your guess for r and start with (a) again.

Compute the interest rate and aggregate savings rate for the economy where $\delta = 0.6, \sigma = 5, \sigma_y = 0.4$.